As of: March 21, 2025
Received: March 15, 2025
Status: [redacted]
Tracking No. m8a-jta0-686e
Comments Due: March 15, 2025
Submission Type: API

PUBLIC SUBMISSION

Docket: NSF_FRDOC_0001
Recently Posted NSF Rules and Notices.

Comment On: NSF_FRDOC_0001-3479
Request for Information: Development of an Artificial Intelligence Action Plan

Document: NSF_FRDOC_0001-DRAFT-2312
Comment on FR Doc # 2025-02305

Submitter Information

Name: Patrick DeMicco

[redacted]

General Comment

See attached file

Attachments

The Role and Possible Challenges Artificial Intelligence Poses to the Authentication and Reliability of Digital Evidence

Re: Request for Information on the Development of an "AI Action Plan"

Submitted to: Networking and Information Technology Research and Development (NITRD) National Coordination Office (NCO), National Science Foundation. ([redacted]

Submitted By: Patrick DeMicco, [redacted]

Date: March 15, 2025

I. STATEMENT OF INTEREST

As a third-year law student at the University of Akron School of Law studying Artificial Intelligence Law and Policy, I appreciate the opportunity to provide input on the development of the AI Action Plan. My growing expertise in artificial intelligence law and policy informs this response which addresses the role and possible challenges artificial intelligence poses to the authentication and reliability of digital evidence.

II. INTRODUCTION: DIGITAL FORENSICS AND ARTIFICIAL INTELLIGENCE IN LEGAL PROCEEDINGS

As AI develops and becomes more integrated into various aspects of life, the legal landscape surrounding it remains largely unexplored. Since AI is still a somewhat new topic, there is a lack of case law that properly addresses its complexities. As a result, lawmakers find themselves navigating uncharted territory. They must rely on current laws and guidelines while they wait for the creation of new legislation specifically designed for AI. The dynamic character of AI poses a range of obstacles and opportunities, making it necessary for the legal system to adapt as it attempts to keep up.

III. EXAMINING LEGAL PRECEDENT: LANDMARK CASES SHAPING THE ADMISSIBILITY OF EXPERT EVIDENCE

Few cases in legal history have had as big of an impact on evidence admissibility as *Frye v. United States*, 293 F. 1013 (D.C. Cir. 1923). Previously, there had been a "sporting theory" in law with the consensus being that courts should let each side have equal opportunity to present their experts and have the judge act as the "umpire".[1] In *Frye*, the court held that the standard for the admissibility of expert evidence is that it must have gained general acceptance in that scientific community.[2] *Frye*, with its decision issued over a century ago, continues to resonate in law to this day, influencing the modern standards by which expert testimony is evaluated.

IV. EXPLORING AUTHENTICATION: NAVIGATING DIGITAL FORENSICS IN LEGAL CONTEXTS AND ASSESSING THE RELIABILITY OF DIGITAL EVIDENCE

A. Case Studies: Digital Forensics in Legal Proceedings

As digital forensics continues to gain importance in the legal world, its impact on cases that involve electronic evidence becomes clearer. The use of digital evidence in court has been

[1] Alfredo Garcia, *The Compulsory Process Clause and the "Sporting Theory of Justice": The Supreme Court Evens the Score*, 28 DUQ. L. REV. 619 (1990).

[2] *Frye*, 293 F. at 1014

influenced by key cases which have changed the field of digital forensics in law, such as defining how expert testimony is to be used, the reliability of digital evidence, and more.

Social media is a relatively new thing. Forty years ago, there was no Facebook, there was no Twitter, there was no Snapchat. Even twenty years ago, early social media such as blogs were still new. But in the modern era, social media is commonplace. Due to this, there have been many court cases where social media plays a crucial role in the legal proceedings.

In re K.B., 238 Cal. App. 4th 989, 190 Cal. Rptr. 3d 287 (2015) is a case that highlights how courts may go about authenticating social media evidence. Officer Ochoa was the "Instagram officer" for the San Francisco Police Department, tasked with scrolling through Instagram to track and monitor individuals.[3] Officer Ochoa, who was already familiar with appellant from prior investigations, saw appellant posing with firearms on Instagram.[4] Officer Ochoa knew appellant was a wanted felon on probation and was prohibited from possessing firearms, so he and his partner decided to perform a probation search.[5] The officers went to the address of appellant and saw the same surroundings as in the Instagram photos.[6] The officers found appellant with firearms and arrested him.[7]

Appellant's cell phones were seized, and the officers found the photos they had seen on Instagram.[8] Officer Ochoa took pictures of each photo on appellant's phone.[9] The officers then conducted a Cellebrite search, which is a type of software commonly used to retrieve information from cellphones.[10] Once a Cellebrite search is performed, an officer can push a button and a report is printed that contains all the cell phone's information such as "photographs; incoming, outgoing, and missed calls; text messages; app information; and e-mails."[11] Included in appellant's Cellebrite report were the same incriminating photos that the officers had seen earlier and taken pictures of.[12]

At trial, over defense counsel's objections, the court admitted into evidence the pictures taken by Officer Ochoa and the Cellebrite report containing the same photographs.[13] Appellant did not testify and rested on the state of the evidence and was sentenced to complete an eight-month program at a juvenile center.[14] Appellant appealed, arguing that admission of the evidence was improper due to improper authentication as there was no testimony that established the photos were actually what they depicted.[15]

On appeal, the appellate court held that the admission of the evidence had been proper.[16] It elaborated that the general rule for authentication of a photo or video is by showing that it is "a

[3] *Id.*
[4] *Id.*
[5] *Id.*
[6] *Id.*
[7] *Id.*
[8] *Id.*
[9] *Id.*
[10] *Id.*
[11] *Id.*
[12] *Id.*
[13] *Id.*
[14] *Id.*
[15] *Id.*
[16] *Id.*

fair and accurate representation of the scene depicted" and reasoned that the foundation for admitting the photos in this case had been properly laid by the investigating officers.[17]

This case emphasizes how crucial strict authentication protocols are for the admissibility of digital evidence. Officer Ochoa's in-depth documentation and use of the Cellebrite program led to appellant raising concerns about the reliability of the incriminating photographs shown during trial.[18] The decision by the appellate court to uphold the evidence's admissibility is consistent with the ever-changing standards needed to address the challenges posed by digital investigations.[19]

With the advancement of technology, the ability to enhance videos also emerged. This technological tool offers users the ability to provide better clarity to low-resolution images. However, this brings up an interesting question – is an enhanced video considered a different piece of evidence under evidentiary standards? This was the question in *State v. Melendez*, 291 Conn. 693, 970 A.2d 64 (2009).

Melendez involved a sting operation, with the Drug Enforcement Agency (DEA) setting up a sting on a known drug operation.[20] The DEA had informant wear a device that captured both audio and visual and enter a restaurant, where he was to buy narcotics from anyone selling them.[21] After entering the restaurant's bathroom, informant encountered defendant, whom he knew to be a drug dealer.[22] Defendant sold informant narcotics, and informant immediately left and met with the DEA, where they got a positive identification of cocaine as well as a positive identification of defendant's identity.[23] Two weeks later, informant did the same thing and again caught defendant on video selling him narcotics.[24]

Based on informant's identification of defendant, defendant was arrested and charged.[25] After viewing the incriminating videos during a pre-trial hearing, defendant refused a plea and decided to go to trial.[26] At trial, there were enhanced versions of the incriminating videos, with some slowed to ten percent of normal speed.[27]

Defendant argued that because the DVD contained video of the two transactions that could be seen more clearly and definitively than the images on the videotapes that he had viewed, the DVD "effectively constituted new evidence that the state previously had not made available to the defendant."[28] Defendant argued that if he had received these versions earlier, he would have taken the plea for a lesser sentence and thus the state should renew that offer.[29]

---

[17] *In re K.B.*, 190 Cal. Rptr. 3d at 289 (quoting *People v. Goldsmith*, 59 Cal. 4th 258, 172 Cal. Rptr. 3d 637, 326 P.3d 239 (2014))

[18] *In re K.B.*, 190 Cal. Rptr. 3d at 289

[19] *Id.*

[20] *Id.*

[21] *Id.*

[22] *Id.*

[23] *Id.*

[24] *Id.*

[25] *Id.*

[26] *Id.*

[27] *Id.*

[28] *Id.*

[29] *Id.*

The trial court denied defendant's motion for a renewed offer, stating that he did not meet his burden of establishing that the state should renew their offer.[30] It reasoned that defendant had "previously had been afforded the opportunity to review the same videotape that the state had used to make the [enhanced] DVD, and that the videotape and DVD depict exactly the same transactions."[31] The court ultimately allowed the state to introduce the DVD into evidence through the DEA agent who had set up the videotape to be enhanced, with that agent testifying that the DVD footage was the exact same as the original tapes' footage except more clear.[32]

On appeal, defendant argued that admission of the DVD was improper because the state failed to lay the proper foundation for its introduction into evidence.[33] The appellate court ultimately held that the DVD's introduction into evidence was proper.[34] It reasoned that although it agrees with defendant that the trial court should not have admitted the portions of the DVD with the enhanced footage, it was not improper for the court to admit the footage as it "merely represented an exact copy of the footage from the original eight millimeter videotape" and that its admission was "harmless beyond a reasonable doubt."[35] The court elaborated that defendant cannot prove that he was harmed by the modified footage's admission into evidence since the clips shown to the jury contained the exact same images as the original clips.[36] The appellate court ultimately affirmed the conviction.[37]

Melendez provides an interesting point of discussion regarding images being admitted into evidence.[38] In these types of cases, where enhanced images are admitted but are basically identical to the original images, any potential error in admitting the enhanced images may not result in harm to the defendant.[39] This highlights a subtle aspect of evidence admission by highlighting the fact that, despite any enhancements, the core content of the evidence is unaltered and thus there is almost never any harm to defendant.

Authentication is the main issue in many cases involving digital forensics. Investigations involving the use of digital forensics can be severely compromised by faulty authentication. Thus, proper authentication is key for a case to go smoothly. One method of authenticating evidence is with expert testimony, where qualified experts examine and verify the authenticity of digital evidence. One case that highlights how crucial expert testimony can be in the authentication of evidence is *Lamb v. State*. 246 So. 3d 400 (Fla. Dist. Ct. App. 2018).

In *Lamb*, defendant was involved in multiple car jackings.[40] The first victim was sitting in his car outside a Jupiter hotel at around 10:00 pm when a man with a gun came up to him and forcibly removed him from the vehicle.[41] The first victim's car, phone, watch, wallet, and cash – including money from Cuba – were taken by the carjacker's accomplice.[42] A similar incident

---

[30] *Id.*
[31] *Id.*
[32] *Id.*
[33] *Id.*
[34] *Id.*
[35] *Id.*
[36] *Id.*
[37] *Id.*
[38] 291 Conn. 693, 970 A.2d 64
[39] *Id.*
[40] *Id.*
[41] *Id.* at 403
[42] *Id.*

happened in Greenacres at around one in the morning.[43] Two attackers, one of whom had a silver gun, robbed the second victim of his possessions and car.[44] The second victim saw a car that looked like the first one drive by, and the drivers spoke to each other before the car took off.[45] The second victim was unable to make out the identity of the driver or whether there were any other people in the car.[46]

Both victims' cars were found in the same area in a city between Jupiter and Greenacres.[47] The phones of the first victim and second victim were discovered in the first victim's vehicle, but the cash, wallet, and watch were gone.[48]

The investigation of the first carjackings was conducted by a Jupiter police detective while the second one was conducted by a Palm Beach County sheriff's detective.[49] Because of the similarities between the two cases, the detectives ended up working together.[50] When brought in by detectives, both victims identified the codefendants, but did not identify defendant.[51] The detectives found connections among the codefendants, who all lived in the same area where the cars were found.[52] When a codefendant's phone was searched, detectives found pictures of him with a silver gun that matched the one used during the carjackings.[53] Police arrested another codefendant with a similar gun.[54] Additionally, a Facebook video showing both stolen cars was discovered by the Jupiter police department on defendant's phone.[55] Prior to trial, investigators showed the first victim the Facebook video.[56]

During the trial, the state showed the Facebook video to the first victim, who identified defendant, and later to a detective, who also identified defendant.[57] The trial court admitted this testimony despite challenges raised regarding the best evidence rule. The video, posted shortly after the second carjacking, showed defendant driving victim's car and wearing victim's watch.[58] A co-defendant was also present in the video.[59]

The state then moved to admit the Facebook video.[60] Defendant objected, and then a sidebar occurred.[61] At the sidebar, defense counsel cited lack of authentication for the Facebook video allegedly extracted by the Jupiter Police Department.[62] Defense emphasized that the sheriff's detective, who testified, did not extract the video but simply viewed it.[63] The state

---

[43] *Id.*
[44] *Id.*
[45] *Id.* at 403-404
[46] *Id.* at 404
[47] *Id.*
[48] *Id.*
[49] *Id.*
[50] *Id.*
[51] *Id.*
[52] *Id.*
[53] *Id.*
[54] *Id.*
[55] *Id.*
[56] *Id.*
[57] *Id.*
[58] *Id.*
[59] *Id.*
[60] *Id.*
[61] *Id.* at 404-405
[62] *Id.* at 405
[63] *Id.*

argued that the video was self-authenticating because it showed defendant saying it is “live” and that it lines up with the timing of the carjackings.[64] The court questioned the method of extraction, insisting that someone must testify to having downloaded it from Facebook for proper authentication.[65]

After the sidebar, the court sustained defendant’s objection based on lack of authentication.[66] The state tried to authenticate the video through a forensic examiner, who testified about his experience and methodology.[67] Despite objections about the examiner’s classification as a witness and concerns about the lack of testimony from someone present during the recording, the court ruled the video was sufficiently authenticated based on its content and the examiner’s testimony.[68] The court admitted the video into evidence along with a screenshot of codefendant’s Facebook page. In the end, defendant was found guilty of grand theft of a vehicle and watch during the first carjacking but was acquitted of charges related to the second carjacking.[69]

On appeal, defendant’s arguments related mainly to what the Facebook video revealed.[70] Defendant alleged that the trial court erred on four points; failure to recognize a discovery violation regarding the digital forensic examiner, admission of the Facebook video without proper authentication, overruling objections to witness identifications in the video, and denial of the motion for judgment of acquittal based on lack of proof of intent in the carjackings.[71]

Regarding the alleged lack of authentication of the Facebook video, the court held that the state met the low threshold required to authenticate the Facebook video since the examiner confirmed the presence of the video on a codefendant’s page, showing defendant driving the stolen vehicles.[72] The video’s authenticity was further supported by the testimonies from the victim and the Jupiter detective confirming defendant’s appearance and actions in the video.[73] Based on this evidence, the appellate court concluded that the state had made a prima facie showing of the video’s authenticity for the purpose of admission into evidence, which would allow the jury to ultimately decide how much weight it should be given.[74]

Defendant brought up *Santana v. State*, a case in which there was an issue regarding audio authentication, suggesting that "authentication should be made by the technician who operated the recording device or a person with knowledge of the conversation that was recorded."[75] However, the appellate court distinguished *Santana* from the matter at hand by stating that the issue in *Santana* was an audio recording which could have been altered without detection but here, the Facebook video “provides an unbroken visual recording of the defendant for an extended period.”[76]

---

[64] *Id.*
[65] *Id.*
[66] *Id.*
[67] *Id.*
[68] *Id.* at 406
[69] *Id.* at 407
[70] *Id.*
[71] *Id.*
[72] *Id.* at 408
[73] *Id.*
[74] *Id.* at 409
[75] *Lamb*, 246 So. 3d at 409 (quoting *Santana v. State*, 191 So. 3d 946 (Fla. Dist. Ct. App. 2016))
[76] *Lamb*, 246 So. 3d at 409 (quoting *Santana v. State*, 191 So. 3d 947 (Fla. Dist. Ct. App. 2016))

In the end, the appellate court held that the district court did not abuse its discretion in admitting the video into evidence over defendant's objection alleging that it was not properly authenticated.[77] Finding no validity in any of defendant's arguments, the appellate court affirmed his convictions.[78]

V. DECODING THE ROLE OF AI: EXAMINING ITS IMPACT ON AUTHENTICATION, RELIABILITY, AND LEGAL PRECEDENTS IN DIGITAL FORENSICS

A. Case Studies: AI

Although AI is a relatively new technological development, it has already started to make its mark in some legal proceedings. This can be seen through emerging caselaw, such as courts ruling that AI cannot be considered the inventor of something.[79]

With AI's increasing prevalence in legal proceedings, one thing worth considering is how the Confrontation Clause will factor into cases that involve AI-generated reports. If an AI program generates a report, how does one confront those against him? Can one confront AI? If so, how does that work? Is it possible for one to confront an AI program?

It has generally been held that AI generated reports are not subject to the Confrontation Clause.[80] In *Weeden*, appellant blocked victim's vehicle from exiting her driveway and approached the driver door carrying a gun.[81] Victim was eventually able to drive off with her daughter and escape, but not before appellant fired his gun at the car, hitting it twice.[82] Appellant was charged with one count each of "aggravated assault, person not to possess a firearm, carrying a firearm without a license, and propulsion of missiles into an occupied vehicle, and three counts of recklessly endangering another person."[83] At trial, both victims testified as well as a detective, who testified for the Commonwealth detailing the Bureau's use of a gunfire detection program, "ShotSpotter."[84]

The detective testified that ShotSpotter is a program designed with the aim of "detecting, triangulating, and pinpointing the location of any loud [noises]" within the city's limits.[85] The program uses sensors throughout the city to detect these noises.[86] The detective testified that when the program detects a loud noise, it automatically documents it and sends it to a human operator in California whose job is to determine whether the noise was a gunshot.[87] If the operator determines that the noise was a gunshot, they will send the data back to the Bureau and officers will be dispatched to that location within seconds.[88] According to the detective, the human operators receive hundreds of hours of training "to be able to recognize the difference between . . . pattern[s] and . . . sound[s]" of gunshots, to the point that they can discern the

[77] *Lamb*, 246 So. 3d at 409
[78] *Id.* at 413
[79] *See Thaler v. Vidal*, 43 F.4th 1207 (Fed. Cir. 2022); holding that under Patent Act, 35 U.S.C.S. § 100(f) only humans can be inventors).
[80] *Commonwealth v. Weeden*, 304 A.3d 333 (Pa. 2023)
[81] 304 A.3d at 335
[82] *Id.*
[83] *Id.*
[84] *Id.* at 336
[85] *Id.*
[86] *Id.*
[87] *Id.*
[88] *Id.*

difference between a firecracker and a gunshot.[89] The detective acknowledged that the system and its operators are not foolproof, but that the ShotSpotter system is "very accurate" in detecting the presence of gunfire.[90]

The Commonwealth then proffered into evidence, via the detective, a "ShotSpotter Investigative Lead Summary,' which is automatically generated by ShotSpotter and provides the date, time, and location of the suspected gunshot.[91] The detective added that after the summary is automatically generated, "updates may be added by the ShotSpotter operators to depict information obtained by the responding police officers."[92]

Relating to the case, the detective testified that officers were dispatched to appellant's location after ShotSpotter detected possible gunfire in that location.[93] In support of his case, appellant testified that he had supported the victim financially during their relationship and that she was unemployed at the time of the shooting.[94] Appellant also presented testimony from multiple witnesses who attested that appellant was with them the evening of the shooting "playing video games and eating soup" and that he did not leave until the following morning.[95]

Ultimately, appellant was found guilty on all charges.[96] Appellant appealed, arguing that the court "violated his right to confrontation under the state and federal constitutions by admitting the Summary into evidence, as, in his view, the Summary was testimonial in nature, such that he should have been afforded the opportunity to cross-examine the declarant who created it."[97] Appellant maintained that his cross-examination of the detective did not suffice because the detective had no role in generating the report.[98] The trial court held that appellant deserved no relief on his constitutional challenge, as he failed to specify an exact person that he wanted to confront.[99] It further held that the creator of the Summary may not be subject to cross-examination because the Summary is computer-generated.[100]

On appeal, the main issue was whether the ShotSpotter Summary is testimonial in nature and therefore subject to the protections afforded under the Confrontation Clause.[101] Appellant maintained that the Summary was testimonial as it was, in his view, offered "to prove the precise time and location of the alleged shooting incident."[102] Appellant also argued that the creation of the Summary contained a large human component due to the provision that stated it is reviewed by a human.[103] Appellant highlighted the detective's testimony where he stated that the reports generally are reviewed by a human operator, but that he was unsure if that process occurred in this case.[104] Nonetheless, appellant asserted that he deserved the right to confront the human

[89] *Id.*
[90] *Id.*
[91] *Id.*
[92] *Id.*
[93] *Id.* at 336-37
[94] *Id.* at 338
[95] *Id.*
[96] *Id.*
[97] *Id.*
[98] *Id.*
[99] *Id.*
[100] *Id.*
[101] *Id.* at 339
[102] *Id.* at 340
[103] *Id.*
[104] *Id.*

operator for ShotSpotter that worked on this case, as well as any forensic engineer that may have reviewed the data.[105] Again, appellant argued that his cross-examination of the detective did not suffice because “he had no role in creating the [Summary] and thus could not speak to its reliability."[106]

The appellate court ultimately held that the Summary was non-testimonial in nature given the fact that the ShotSpotter system “recorded the data in an effort to assist law enforcement in responding to an ongoing emergency.”[107] It reasoned that the data in the Summary was not generated with the primary purpose to establish or prove past events relevant to a later criminal prosecution, but rather it was “created to aid law enforcement in combating a particular instance of gun violence and, more specifically, to permit the Bureau to immediately dispatch officers to the . . . location to investigate and discern whether shots had been fired.”[108] Thus, the admission of the report did not violate the Confrontation Clause or *Crawford*, and the court rejected appellant’s evidentiary challenges to the Summary.[109]

Weeden highlights an interesting subsection of AI and its uses, that being the admissibility of AI-generated data and reports.[110] The case highlights the fact that these types of reports are typically not considered hearsay because they are automatically generated by a program rather than being asserted by a person.[111] It is also worth noting that reports such as the ShotSpotter Summary are not subject to the Confrontation Clause for the same reason; it was generated by a system rather than a person[112][113]

AI is constantly evolving and improving. Given its rapid development and capabilities, it is no surprise that AI has been used to calculate people’s reactions to things. This was the topic of *Bertuccelli v. Universal City Studios LLC*, No. 19-1304, 2020 U.S. Dist. LEXIS 195295 (E.D. La. Oct. 21, 2020).

Bertuccelli involved using AI to calculate people’s reactions to two similar masks.[114] In 2009, plaintiffs created King Cake Baby (“KCB”) and licensed its image for use as the mascot of the New Orleans Pelicans.[115] In 2017, the film “Happy Death Day” released in theatres, in which a serial killer wears a cartoon baby mask that bears a noticeable resemblance to plaintiff’s KCB.[116] Plaintiffs argue that defendant’s film and its sequel infringe on their KCB copyright. [117]

Defendants moved to exclude any opinion testimony by plaintiff’s experts – Mr. Berger

---

105 *Id.*

106 *Id.*

107 *Id.* at 350

108 *Id.* at 351

109 *Weeden*, 304 A.3d at 351; *see Crawford v. Washington,* 541 U.S. 36, 124 S. Ct. 1354 (2004); holding that testimonial statements of witnesses not present at trial admissible only where declarant unavailable and defendant had prior opportunity for cross-examination

110 304 A.3d at 350-51

111 *Id.* at 356-57

113 *Weeden*, 304 A.3d at 350-51; *see People v. H.K.*, 2020 NY Slip Op 20232, 69 Misc. 3d 774, 130 N.Y.S.3d 890 (Crim Ct.); holding AI used as tool to assist analyst in interpretation of data and was not working independently; defendant’s right to confrontation was not violated as analyst was responsible for the AI results, so analyst could be cross-examined

114 *Id.*

115 *Id.* at *2

116 *Id.*

117 *Id.*

and Dr. Griffor – as inadmissible under the Federal Rules of Evidence and *Daubert*.[118] In addition to alleging that the experts' testimony is inconsistent with the court's history of proving "substantial similarity" in copyright cases, defendants also argued that the testimony is unreliable and that neither witness is qualified to opine on issues related to copyright infringement.[119]

The court held that both experts were qualified to testify as experts.[120] It found that Mr. Berger's expertise in evaluating intellectual property perception and conducting trademark surveys established him as a credible witness.[121] Regarding Dr. Griffor, the court held that he is a recognized expert in facial recognition technology, being the Associate Director for Cyber-Physical Systems at the National Institute of Standards and Technology (NIST) in Washington, DC and with qualifications from the Massachusetts Institute of Technology and the University of Oslo.[122]

The court ultimately denied the motion to exclude the testimony.[123] However, it did note that plaintiffs were free to explore their concerns with the experts on cross-examination.[124]

Bertuccelli is an interesting case, as one of the court's holdings was that Dr. Griffor was a qualified expert, citing his conduction of AI-assisted facial recognition analysis as "reliable methodology."[125] The case highlights a common concern regarding expert witnesses, which is their competence. Because expert witnesses can have a big impact on how cases turn out, there is often concern about their qualifications when testifying in court. One aspect of this concern centers on the credentials and experience of the expert. Expert witnesses are usually required by the courts to have specific knowledge, expertise, training, experience, or education related to the subject matter of their testimony.

Another important factor is the reliability of the expert's methods. Courts frequently assess whether the expert's methods have been appropriately applied to the case at hand, and whether they are generally accepted within the relevant field. Preserving the integrity of the trial process requires making sure that the expert's opinions are founded on reliable scientific or technical principles.

In *Bertuccelli,* the court found that both experts had sufficient credentials and experience to deem them qualified.[126] It also found that both experts' methodologies were reliable and credible.[127] This is noteworthy because of how new AI is; considering it has just become widely available in recent years, a court holding that one's methods of using AI are reliable and credible may provide insight into what credentials AI experts should strive to have. This case may set precedent for future instances where courts must decide whether AI experts are qualified.

B. Navigating a New Frontier: AI's Impact on the Legal Landscape

1. AI Deepfakes

[118] *Id.*; *Daubert*, 43 F.3d at 1311
[119] *Id.* at *3
[120] *Id.* at *5
[121] *Id.*
[122] *Id.* at *5-*6
[123] *Id.*
[124] *Id.*
[125] *Bertuccelli*, No. 19-1304, 2020 U.S. Dist. LEXIS 195295 at *6
[126] *Id.*
[127] *Id.* at *5

One area of AI that is concerning are deepfakes. A deepfake is an image or recording that "has been convincingly altered and manipulated to misrepresent someone as doing or saying something that was not actually done or said".[128] Deepfakes have been around for years, but up until recently one would have to manually edit something themselves to make one. Thanks to AI, deepfakes are now created faster and look more realistic.

Deepfakes are not limited to visuals, as there exists technology that can create audio deepfakes too. Using AI, audio deepfake technology analyzes audio data and identifies patterns and traits of a target voice, which it then replicates to produce a clone that programmers can use via text-to-speech to say whatever they want.[129] Audio deepfakes pose significant concerns in a legal context due to the potential for them to distort the truth. Due to the ability of these advanced technologies to create convincing audio recordings of people saying things they never did, it can be difficult to tell what is real and what is fake. There are already documented incidents involving audio deepfakes, such as a Baltimore school principal being accused of making racist remarks that he never actually said.[130] Additionally, the rapper Drake made headlines recently after he used AI to sound like Tupac on a new song.[131]

In a legal context, audio deepfakes could be used to fabricate false confessions, impersonate people, and manipulate audio recordings of phone calls, just to name a few possible uses of concern. The potential misinformation caused by audio deepfakes could lead to an undermining of the trustworthiness of legal proceedings.

The ability of deepfaked photos, videos, and audio to deceive and manipulate evidence poses significant evidentiary dangers. The loss of confidence in visual evidence is one of the main concerns, since deepfakes can realistically change people's appearances and behaviors, making it difficult to distinguish real content from fake, deepfaked content. This same loss of confidence would occur in audio evidence due to the ability of audio deepfakes to mimic a person's voice, thus undermining the reliability of recorded audio in legal proceedings.

One aspect of this concern is the fact that deepfakes can be used to fabricate events and give false alibis. With deepfakes, criminals could fabricate convincing but entirely fake scenarios that may be used as evidence in court by superimposing people's faces onto bodies or fabricating false recordings. These deepfakes could be weaponized to frame innocent people as well as

---

[128] Deepfake definition & meaning, MERRIAM-WEBSTER, https://www.merriam-webster.com/dictionary/deepfake (last visited Apr 15, 2024).

[129] Audio deepfakes: Cutting-edge tech with cutting-edge risks, BRADLEY, https://www.bradley.com/insights/publications/2024/01/audio-deepfakes-cutting-edge-tech-with-cutting-edge-risks (last visited Apr 14, 2024).

[130] Lauren Leffer, AI AUDIO DEEPFAKES ARE QUICKLY OUTPACING DETECTION SCIENTIFIC AMERICAN (2024), https://www.scientificamerican.com/article/ai-audio-deepfakes-are-quickly-outpacing-detection/ (last visited Apr 14, 2024).

[131] Bill Donahue, TUPAC SHAKUR'S ESTATE THREATENS TO SUE DRAKE OVER DISS TRACK FEATURING AI-GENERATED TUPAC VOICE BILLBOARD (2024), https://www.billboard.com/pro/tupac-shakur-estate-drake-diss-track-ai-generated-voice/ (last visited Apr 25, 2024)

exonerate guilty people. Through the manipulation of evidence, perpetrators could falsely exonerate or incriminate whoever they want.

Additionally, deepfakes would lead to an increase in doubt of the authenticity of legitimate evidence. Picture this: you are a defense attorney, and you submit a legitimate video of your client at home the night he allegedly committed a robbery. Your client was relaxing with family and decided to snap a photo of him and his mom. The opposing side may reject this photo as a deepfake, raising questions about its legitimacy and reliability even though it accurately depicts what happened. This could lead to your client being unfairly discredited and wrongly convicted, despite presenting genuine evidence that supports his innocence. The mere existence of deepfake technology will lead to skepticism and uncertainty when it comes to legal proceedings, which will make it increasingly difficult to trust the authenticity of evidence. The integrity of the entire legal system would be at risk because of this erosion of trust, which threatens the core values of justice and fairness.

Even outside of a legal context, deepfakes pose significant concerns. With manipulated content, individuals may find themselves depicted in defamatory situations without their knowledge or consent. This could cause significant damage to their reputation and may even have legal repercussions. Defamatory instances of deepfakes have already happened with revenge porn made with AI deepfake technology, as well as child porn made with the same technology.[132]

2. Legislative Responses to Address AI Risks

Given that AI comprises a wide range of technologies and applications in different areas, a diverse range of regulations suited to the various contexts in which AI functions are required. While some bills attempting to regulate AI have already been passed, others are still in the legislative process.

To combat concerns related to AI, many states have started to propose new legislation. New York introduced a bill with the aim of amending the law and criminalizing the unlawful publication, dissemination, and access of sexually explicit depictions of individuals and "establishing offenses involving sexually explicit digital alterations."[133] The law would make committing these actions a class E felony.[134] It is noted in the bill's justification section that as more advanced technology becomes available, child pornographers can circumvent the law and generate child porn using AI.[135] It noted that these digitally manipulated images lead to the "same reputational harm caused by conventional forms of child pornography" but that they are not criminally prohibited in New York."[136] This bill aims to allow courts to finally punish those who have been able to create child porn using AI and avoid facing punishment.[137] As of April 14, 2024, this bill had not yet been passed.[138]

---

[132] Alexis Santiago, *COMMENT: The Internet Never Forgets: A Federal Solution to the Dissemination of Nonconsensual Pornography*, 43 Seattle U. L. Rev 1273-98 (2020).

[redacted] [redacted] Legis. Bill Hist. NY A.B. 7519

[134] *Id.*

[135] *Id.*

[136] *Id.*

[137] *Id.*

[138] *Id.*

At a national level, there was a bill introduced in the House which aims to "protect national security against the threats posed by deepfake technology."[139] It also aims to provide legal recourse to those who have been victims of harmful deepfakes.[140] Currently, it also has not yet been passed.[141]

3. Exploring Non-Legislative Approaches to Address AI Risks

There are other avenues besides legislation that can hopefully counteract the dangers posed by AI. One possible solution to address these concerns regarding AI and authentication is to create special jury instructions that address the issue, specifically when it comes to AI-generated evidence. Pennsylvania courts recently proposed jury instructions to address computer-generated animation evidence.[142] The jury instructions explain that a computer-generated animation will be presented as evidence, and that animation was created by a specific artist based on information provided to them by one of the parties.[143] The instructions also state that the animation does not contain any conclusions or calculations made by the computer itself.[144] In the instructions, it is emphasized that the animation represents that party's recollection of events and is not perfectly representing what happened.[145] The jury is reminded that they are free to accept or reject any part of the animation, just as they can with other evidence.[146] The instructions state that to assess the animation, the jury should consider the credibility of the information that the artist relied on as well as their skill, experience, and education.[147]

AI technology frequently uses complex data analysis techniques and algorithms. The ordinary juror is not likely to fully understand the AI evidence they are being shown. By giving the jury brief and straightforward explanations of AI evidence, the judge would help ensure that jurors understand the nature and significance of the evidence presented.

The jury would be given instructions by the judge to assess the authenticity of the AI evidence by taking into account many things, such as origin and reliability of the data used to feed the AI, the qualifications and experience of those in charge of creating and applying the AI, as well as any known restrictions related to the AI employed in the case. Regarding the weight of the evidence, the judge would indicate that the jury is free to accept or reject any part of the AI evidence submitted by either party. Giving the jury these instructions at the beginning of the trial may help frame what they should be on the lookout for, helping them ultimately form a better verdict.

We have already seen in *Bertuccelli* how the court considered the qualifications of plaintiff's experts when determining that they were qualified.[148] This consideration of experts' qualifications is a small step to better results in legal proceedings involving AI. If a court were to apply jury instructions like I propose, there may be an even better handling of AI evidence in the

---

[redacted] [redacted] H.R. 5586; 118 H.R. 5586
140 *Id.*
141 *Id.*
142 Pa. SSJI (Civ) 3.90
143 *Id.*
144 *Id.*
145 *Id.*
146 *Id.*
147 *Id.*
148 *Bertuccelli,* No. 19-1304, 2020 U.S. Dist. LEXIS 195295 at *5-*6

future. These instructions would help inform the jury on how to view the evidence and the weight to give it, likely leading to better results.

Changes to the Federal Rules of Civil Procedure may be necessary, especially when it comes to discovery rules. As AI systems generate documents, communications, and legal analyses, the scope of what's discoverable may need to be clarified. Courts will have to decide whether AI-generated outputs are subject to discovery, as well as if the underlying algorithms or training data should be discoverable. This may be especially important in cases where possible bias or decision-making processes are at issue.

Additionally, there are many people who believe that there is no need for a law aimed specifically at AI, as almost every situation where AI may arise would already be covered under an existing law. For example, issues with AI when it comes to appropriating someone's likeness would already be covered under the right of publicity, which is a state-law tort designed "to prevent unauthorized uses of a person's identity that typically involve appropriations of a person's name, likeness, or voice."[149] This means that in cases involving the use of AI to replicate a celebrity's voice or image without their consent, that celebrity would have the ability to take legal action under this right of publicity.

VI. Taking the Next Step: Safeguarding Legal Integrity in the Era of AI

Ultimately, my investigation into the complex area where AI meets legal proceedings has been both informative and unsettling. Although AI can be very helpful in various aspects of life, the possible risks that it poses to the credibility of evidence cannot be ignored. I am unsure where exactly AI will lead us in legal proceedings in the future, but we cannot hope to reduce the risks posed by AI without first considering the dangers discussed in this paper. Adopting a myopic attitude towards the dangers of AI will get us nowhere; we must be proactive and take steps to address these risks.

To protect the justice and truth that our legal system is built on, courts must implement rules that guarantee the responsible use of AI. New legislation must also be passed to combat the dangers posed by AI. As we stand on the brink of a future in which AI will be everywhere, we are faced with the important duty of upholding the legal system's integrity in an age of rapid technological advancement.

[149] state-law tort designed to prevent unauthorized uses of a person's identity that typically involve appropriations of a person's name, likeness, or voice